

DANYAL AHMED

Boston, MA | (301) 676-7860 | danyalsmail@gmail.com | linkedin.com/in/danyalahmeddev
github.com/theonlynanu | danyalahmed.dev

U.S. Citizen | Pending Secret Clearance

EDUCATION

Northeastern University, Khoury College of Computer Sciences

Boston, MA

Master of Science in Computer Science | GPA: 3.83 / 4.00

Sept 2024 – Expected Dec 2026

Coursework: Machine Learning, Computer Vision, Natural Language Processing, Foundations of Artificial Intelligence, Database Management Systems, Computer Systems / Operating Systems, Object-Oriented Design, Algorithms & Data Structures

University of Maryland

College Park, MD

Bachelor of Economics, Minor in Astronomy | GPA: 3.32 / 4.00 | College Park Scholar, Life Sciences

2019 – 2022

Certifications: AWS Certified Cloud Practitioner (Feb 2024) | Google IT Support Professional Certificate (Feb 2024)

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C, Bash, Java, C++

Infrastructure & Cloud: Kubernetes (k3s, OpenShift), Helm, Docker, GitLab CI/CD, GitHub Actions, AWS (EC2, S3, Lambda), Google Cloud (Cloud Run, Pub/Sub, Artifact Registry), Vercel, JFrog Artifactory, Linux, NFS

Machine Learning & Data: PyTorch, scikit-learn, NumPy, Pandas, Gymnasium, Matplotlib, Grad-CAM

Backend & Databases: FastAPI, Flask, PostgreSQL, MySQL, REST API design, JWT / OAuth (Okta SSO), Pytest, Postman

Frontend: React, Next.js, Vite, Tailwind CSS, Node.js, server-side rendering

Tooling & Practices: Git, uv, MkDocs, monorepo management, technical documentation

EXPERIENCE

Johns Hopkins University Applied Physics Laboratory

Laurel, MD

Software Engineering Intern

May 2026 – Aug 2026

Navy Mission-Planning Framework — Core Infrastructure Team

- Built the platform's first end-to-end local run system, cutting the edit-to-verify loop from ~15 minutes of push-and-wait GitLab CI cycles to 2–4 minutes on a developer machine, while replacing the CI smoke-test job entirely (~800 lines of pipeline YAML reduced to ~120)
- Automated the full cluster lifecycle in a single command: k3s provisioning, local image build and import, namespaced Helm deployment of core services and third-party workloads, health checks, and trapped port-forward cleanup
- Rewrote a ~2,000-line Bash orchestrator in Python to improve maintainability and remove an Ubuntu/RHEL dependency, exposing ~20 CLI flags across pipeline-augmentation and short-circuit modes (namespace teardown, deep image cleanup, remote image build/push, test-only reruns, stage skipping) with per-stage OK/FAILED/SKIPPED reporting and verbose and silent output modes
- Drove adoption to ~30 engineers across the UI team, classified-environment deployment teams, and three partner organizations, establishing the tool as the canonical way to run the platform in local, CI, and production-image configurations
- Authored Helm charts including parent charts provisioning shared cluster resources such as NFS-backed storage for externally-owned workloads, and decoupled the local k3s path from the production OpenShift cluster so development required no dependency on a classified network
- Designed a reusable adapter framework translating between the core system and external tooling, reconciling incompatible geospatial representations (0.1° lat/lon cells vs. kilometer-based grids) and sensor models (azimuth vs. cone height and radius); shipped one production adapter and the framework now backing a second
- Rebuilt the Pytest integration suite, replacing startup-only coverage with assertions across every run stage that validated API contracts and queried the database directly to confirm state propagation between pipeline phases
- Overhauled GitLab CI by adding stages, partitioning runners by test class, retaining artifacts for historical test comparison, and rewriting trigger rules (latest-commit-only, manual end-to-end gating on main) to eliminate runner contention
- Restructured 30+ pages of MkDocs documentation published to an internal site, adding a glossary and reference section that did not previously exist; program-area management cited the introductory material as substantially more readable
- Led the program's first technical demo to ~25 stakeholders including two program area managers and Navy representatives, running live end-to-end tests and custom visualizations of infrastructure progress
- Coordinated with three external teams on API design, deployment harnesses, and mock data for integration testing, and onboarded two engineers over two weeks with design documentation and diagrams built for continuity after departure

Self-Initiated Staffing Coordination Platform — Funded Proof of Concept

- Identified a manual coordination burden spanning ~75 sections across 12 groups, authored a funding proposal, and pitched it to group leadership to secure dedicated labor hours to build a prototype
- Built a containerized nightly sync in Python ingesting ~1,100 employee and ~90 project records from an internal REST BI service and an HTML-scraped internal site into PostgreSQL, completing a full sync in ~10 minutes and incremental runs in under 2 minutes
- Delivered a Next.js and TypeScript frontend over a FastAPI service layer, authenticated through internal Okta SSO with JWTs issued and validated by an internal API gateway
- Projected a minimum \$15K in annual overhead savings under the most conservative assumption of one hour saved per section supervisor per year, against under \$2K of prototype labor and effectively zero incremental infrastructure cost; section chiefs independently estimated 30–40 hours saved each
- Documented architecture, design rationale, and roadmap to enable handoff; the repository drew six developer access requests and is being transitioned to a full-time engineer

Northeastern University

Boston, MA

Graduate Teaching Assistant, Data Structures & Algorithms

Aug 2025 – Dec 2025

- Led a seven-student section with one-on-one coaching in C programming, algorithmic analysis, and systems concepts, receiving a favorable end-of-term review from the supervising professor
- Guided debugging, optimization, and runtime performance analysis on assignments spanning divide-and-conquer, dynamic programming, and graph algorithms
- Designed and reviewed instructional material on manual memory management, Git workflows, and asymptotic reasoning, and expanded office hours in response to sustained student demand

Micro Center

Cambridge, MA

Customer Service Representative

Oct 2024 – May 2026

- Advised customers on hardware specifications and performance trade-offs, translating technical constraints into recommendations for non-technical buyers
- Handled returns, service intake, point-of-sale systems, and inventory reconciliation while working part-time alongside full-time graduate coursework

Magoo's Pub & Eatery

Frederick, MD

Manager

2018 – 2024

- Supervised teams of up to six staff serving 200+ guests daily while holding comps and voids under 1% of sales
- Designed training and onboarding processes for 20+ new employees, and managed sales accounting and vendor communication as keyholder and daily lead

RESEARCH

Galaxy Morphology Classification Under Observational Domain Shift | Python, PyTorch, scikit-learn

Spring 2026

- Quantified out-of-distribution degradation across three model families (linear SVM, gradient-boosted trees, and a 1.87M-parameter CNN) on 178K labeled GalaxyZoo2 images partitioned by redshift, with the CNN reaching 0.77 in-distribution and 0.67 out-of-distribution macro-F1
- Designed a controlled bin-matched comparison isolating domain membership from redshift effects, showing the CNN lost less than half the macro-F1 the SVM lost at identical redshift
- Identified a cross-model over-prediction artifact via precision-recall decomposition, demonstrating that apparent per-class accuracy gains masked a 32% misclassification rate absorbing true positives from an adjacent class, a failure invisible under aggregate accuracy
- Built a reproducible twelve-stage pipeline (labeling, stratified splitting, feature extraction, CNN and classical training, calibration and domain-shift evaluation, Grad-CAM) with seed-controlled runs, per-run config and metric logging, and a comparative evaluation harness serving both studies from a single codebase
- Engineered a 207-dimensional feature representation (concentration, asymmetry, Gini, M20, radial flux profiles, spatial pyramid gradient histograms) for the classical models, extracted across a 240K-image corpus with a multiprocessing worker pool

github.com/theonlynanu/GalaxyZooClassification

Training CNNs with Human Disagreement: Hard vs. Soft Label Regimes | Python, PyTorch

Spring 2026

- Trained two identically-configured CNNs differing only in loss function (cross-entropy on majority-vote labels vs. KL divergence on vote-fraction soft targets) on a 50K-image stratified subset, holding architecture, hyperparameters, seed, and data splits fixed to isolate loss-driven effects
- Measured a 3.4 percentage-point top-1 accuracy gain under soft-label training and demonstrated it was concentrated almost entirely at one class boundary rather than distributed across the task
- Evaluated calibration using expected calibration error, Brier score, reliability diagrams, and ambiguity-stratified subsets, surfacing systematic underconfidence on the soft-label model's highest-accuracy classes

- Traced an apparent aggregate calibration result to a class-composition artifact through per-class stratification, and generated Grad-CAM attention maps to compare model behavior on ambiguous samples
- Implemented the soft-target objective explicitly rather than using PyTorch's built-in KL divergence to avoid NaN propagation from zero-valued targets in unreached decision-tree branches

PROJECTS

ITMS — Issue Tracking Management System | Flask, MySQL, React, TypeScript, Vite Fall 2025

- Built a full-stack issue tracker with a 31-endpoint Flask REST API that owns all authorization, so role permissions cannot be bypassed by direct API calls or browser tooling
- Designed a normalized eight-table MySQL schema with foreign keys, ENUM domains, regex check constraints, and cascade rules chosen per relationship, allocating per-project sequential issue numbers in a stored procedure guarded by a composite unique constraint on (project_id, issue_number)
- Implemented append-only issue history via AFTER INSERT/UPDATE triggers performing NULL-safe field comparison to log old and new values per changed field, enabling full reconstruction of an issue's state over time
- Enforced authentication through server-side sessions over HTTP-only cookies rather than client-stored tokens, keeping session invalidation under server control

github.com/theonlynanu/issue-tracker

Q-Learning Agents in Pac-Man | Python, Gymnasium, NumPy, reinforcement learning Fall 2025

- Built a CLI-driven training harness exposing epsilon schedule, discount factor, decay rate, and checkpoint frequency as tunable parameters, with policy serialization that preserves state-action visit counts to enable curriculum-style progressive retraining
- Designed a compact relative state abstraction reducing a theoretically intractable state space to 278K discrete states while remaining agnostic to map size, and structured an eight-stage curriculum to overcome reward-signal variance that stalled early training
- Diagnosed persistent policy oscillation and resolved it with a heading-relative action space, then quantified the resulting fourfold state-space expansion as evidence that tabular methods could not scale to the domain
- Concluded from an average of 1.88 visits per explored state that sparsity, not hyperparameter tuning, was the binding constraint, and recommended DQN and CNN-based approaches in a written technical report

github.com/theonlynanu/QPacman

danyalahmed.dev — Personal Site & Portfolio Infrastructure | Next.js, TypeScript, FastAPI, Docker, GCP 2023 – Present

- Built and maintain a Next.js/TypeScript portfolio site scoring 100 Lighthouse performance on desktop, structured with a git submodule separating the Vercel deployment repository from a superproject holding the API and full development history
- Containerized and deployed a game-AI opponent to Google Cloud Run rather than co-locating it in the Next.js app, to work hands-on with container builds, image registries, and managed cloud deployment
- Reassessed the design after evaluating the workload, concluding the state space did not justify a network round-trip; currently migrating the opponent to an in-browser minimax implementation

github.com/theonlynanu/personal-site

LEADERSHIP & COMMUNITY

- Eagle Scout, Troop 1812 — National Capital Area Council
- Volunteer, Frederick Memorial Hospital — 200+ hours of patient transport and administrative support